

THERMAFORM INDUSTRIAL SYSTEMS INC.

ThermaPress Pro 2000

Industrial Thermal Press Service Manual

2000 kN Precision Hydraulic Compression & Molding Press | Model TPP-2000

Document Identification:	MAN-TPP2000-SVC-V3
Applicable Equipment:	ThermaPress Pro 2000 (TPP-2000-H2)
Platen Heating System:	Multi-Zone Dual Electric Platens (up to 350°C)
Publication Date:	November 2025 (Annual Engineering Review)
Service Classification:	Factory Maintenance Personnel & Certified Hydraulic Technicians

Section 1.0: Safety Protocols & Thermal Hazards

The ThermaPress Pro 2000 combines extreme clamping forces (2,000 kN / 200 Metric Tons) with high temperature platens operating continuously at temperatures up to 350°C (662°F). Hydraulic system working pressure reaches 210 bar (3,045 PSI). Strict safety protocols are mandatory.

Thermal Burn Prevention: Platens retain hazardous residual heat for up to 180 minutes after electrical shutdown. Technicians must wear rated Kevlar high-temperature gloves and thermal face shields when working near open platens. Verify platen temperature is under 50°C with an infrared thermometer prior to touching heater wiring or thermocouple blocks.

High-Pressure Fluid Injection Hazard: Hydraulic fluid under 210 bar pressure can penetrate human skin and cause severe tissue necrosis. Never inspect hydraulic leaks with bare hands; always pass a piece of clean cardboard across suspected joints.

Mechanical Ram Safety Lock: Prior to entering the daylight area between platens, mechanically engage the hydraulic ram safety drop-arrest latch SL-1 and insert the certified mechanical safety bar.

Section 2.0: Machine Specifications & Hydraulic Circuits

The ThermaPress Pro 2000 utilizes an electro-hydraulic proportional control architecture with regenerative ram advance and precision platen temperature uniformity.

Key Technical Ratings:

Subsystem Parameter	Rated Value	Operating Range
Maximum Clamping Force	2000 kN (200 Tons)	Proportional 100 - 2000 kN
Platen Dimensions	1000 mm x 1000 mm	Chrome-plated alloy steel
Maximum Temperature	350°C (662°F)	Uniformity ±1.5°C across zone
Hydraulic Fluid Type	ISO VG 46 Anti-Wear Fluid	Reservoir Capacity: 120 Liters
Max Hydraulic Oil Temp	65°C (149°F) Max Threshold	Optimal Band: 40°C - 50°C
Thermocouple Sensor Type	Type-K Dual Mineral Insulated	Nominal: 10 - 15 Ω at 25°C

Section 2.4: Alarm Code Diagnostic Matrix

The machine controller continuously checks sensor telemetry. When a safety or process threshold is breached, the relevant code is registered in the event log and displayed on the front console.

Code	Alarm Name	Trigger Threshold	Primary Section
E101	Platen Temperature Sensor Circuit Open / Thermal Runaway Lockout	Thermocouple Loop Resistance > 10 MΩ	Section 3.1 (Page 5)
E104	Hydraulic Ram Main Relief Pressure Exceeded	System Pressure > 225 bar for 500 ms	Section 4.1 (Page 7)
E205	Proportional Valve SV-1 Position Drift	LVDT Feedback Deviation > 5%	Section 4.2 (Page 7)
E302	Chamber Vacuum Seal Degradation	Vacuum Level Under 50 mbar	Section 6.0 (Page 10)
E410	Safety Light Curtain Beam Interrupted	Optical Receiver Channel Break	Section 1.2 (Page 2)

Notice on Thermal Alarms: Alarm E101 disconnects contactor K1 to avoid uncontrolled heating. Always treat an E101 alarm with high priority.

Section 3.1: Thermal System Faults & Platen Diagnostics

Error E101: Platen Temperature Sensor Circuit Open / Thermal Runaway Safety Lockout

Error Meaning: Error E101 indicates an open circuit or infinite electrical resistance condition detected on the upper or lower platen Type-K thermocouple sensing loops, triggering an immediate emergency heating element lockout to prevent unmonitored platen thermal runaway.

Probable Causes:

1. Snapped or disconnected Type-K thermocouple lead wire at junction box JB-2.
2. Fractured mineral-insulated thermocouple probe sheath resulting from mechanical vibration or platen movement.
3. Loose or oxidized terminal screws on terminal strip TB4-12.
4. Welded solid-state relay (SSR) contact holding heater circuit continuously energized.

Step-by-Step Corrective Action:

1. Turn off platen heating master circuit breaker CB3 and allow platens to cool below 50°C before opening junction enclosures.
2. Disconnect thermocouple leads and measure resistance across terminals T1+ and T1- using an ohmmeter. A healthy sensor measures 10 to 15 ohms at 25°C ambient temperature. A resistance reading greater than 10 Megohms confirms an open or snapped thermocouple probe.
3. Inspect terminal strip TB4-12 and re-torque all wire clamping screws to 0.6 Nm using an insulated torque screwdriver.
4. Check the status indicator LED on solid-state relay SSR-1; if the LED remains illuminated while the controller heating command is OFF, the SSR is internally shorted and welded.
5. Replace failed Type-K thermocouple assembly with Part #TH-2000-K or replace damaged solid-state relay with Part #SSR-75A.

Escalation Procedure (If Initial Corrective Action Does Not Fix It):

If the thermocouple probe measures within nominal range (10 - 15 ohms) and wiring continuity is verified back to cabinet C1, but Error E101 persists immediately upon power-up, the cold-junction compensation circuit on the main temperature controller module has failed. Replace platen controller module Part #TC-MOD-01 on DIN rail 3.

Section 3.2: Platen Heating Zones & SSR Switching

Platen thermal distribution is managed across four distinct PID zones (Top Center, Top Edge, Bottom Center, Bottom Edge). Each zone is switched at zero-crossing AC line voltage by 75A solid state relays mounted on aluminum heatsinks in enclosure C1.

Thermocouple Extension Lead Rules: Only use Type-K compensating extension wire (yellow jacket: positive chromel, red jacket: negative alumel). Never splice copper wire into thermocouple runs, as this introduces parasite thermocouple EMF junctions causing temperature reading offsets of 20°C to 50°C.

Section 4.0: Hydraulic Clamping System Diagnostics

Section 4.1: Error E104 - Hydraulic Ram Pressure Relief Trip

Meaning: Pressure transducer PT-1 detected cylinder pressure exceeding 225 bar. Check proportional relief cartridge RV-1 pilot orifice for metal particle debris.

Section 4.2: Error E205 - Proportional Valve Drift

Meaning: Proportional directional valve SV-1 spool feedback LVDT reports > 5% position error from commanded setpoint. Calibrate spool zero point via amplifier card potentiometer P2.

Section 5.3: Hydraulic Power Unit Overheating & Thermal Imbalance

Symptom: Why is Machine B / ThermaPress Pro 2000 Overheating?

Meaning & Symptom Description: Thermal overload and persistent hydraulic oil overheating where reservoir bulk oil temperature exceeds 65°C (149°F) during normal operational pressing cycles, operating without an active E-code or preceding high-temperature alarm trips.

Probable Causes:

1. Mineral scale and silt clogging inside the water-oil shell-and-tube heat exchanger (Unit HE-1).
2. Main hydraulic recirculation unloader valve RV-3 mechanically jammed in the bypass seat, forcing continuous high-pressure pump discharge over the relief valve.
3. Viscosity breakdown and chemical oxidation of ISO VG 46 anti-wear hydraulic oil due to extended service hours.
4. Cooling water proportional solenoid valve SV-2 seized in the closed position or defective 24VDC actuator coil.

Step-by-Step Corrective Action:

1. Measure the cooling water temperature differential between heat exchanger inlet port CH-1 and outlet port CH-2 using an infrared pyrometer. A delta T below 4°C indicates severe tube fouling (nominal delta T is 8°C to 12°C).
2. Isolate the cooling water loop and flush the heat exchanger tube bundle with a 10% sulfamic acid descaling solution for 45 minutes, then flush thoroughly with clean neutral water.
3. Check electrical coil resistance across solenoid valve SV-2 terminals; nominal resistance must measure 32 ohms (± 2 ohms). If resistance reads 0 ohms or open circuit, replace the 24VDC coil.
4. Check oil reservoir fluid level on sight gauge LG-1 and inspect oil clarity; if oil has turned dark brown or emits an acrid burnt odor, drain and refill reservoir with 120 liters of fresh ISO VG 46 hydraulic fluid.

Escalation Procedure (If Initial Corrective Action Does Not Fix It):

If the heat exchanger is verified clean, water valve SV-2 opens fully, and oil is fresh, but oil temperature continues to climb past 65°C within 30 minutes of operation, the main variable-displacement axial piston pump (Pump P1) internal bronze valve plate has experienced severe scoring, bypassing high-pressure fluid directly into the pump case drain. Measure case drain flow into an external graduated cylinder; if drain leakage exceeds 4.5 liters per minute at 210 bar idle, replace Pump P1 Assembly Part #HP-2000-P1.

Section 5.4: Hydraulic Cooling Loop & Heat Exchanger Flow Diagram

The schematic below illustrates the fluid flow paths between the 120L reservoir, Shell-and-Tube Exchanger Unit HE-1, and Proportional Water Solenoid Valve SV-2.

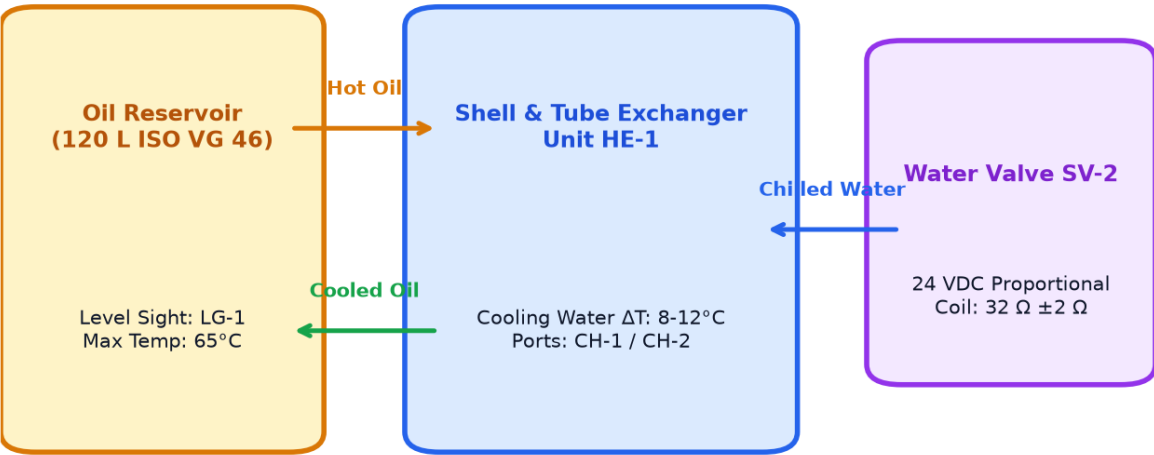


Figure 5.3: Hydraulic Cooling Loop & Heat Exchanger Flow Diagram

Operational Thermal Warning: Running the press with bulk oil temperatures exceeding 65°C degrades cylinder polyurethane piston seals (causing internal leakage) and halves pump bearing service life.

Section 6.0: Vacuum Degassing & Seal Troubleshooting

Section 6.1: Error E302 - Vacuum Seal Degradation

Meaning: Vacuum degassing hood failed to reach 50 mbar within 45 seconds of platen closure.

Corrective Action: Inspect silicone perimeter seal for cuts or cured resin flash. Replace vacuum pump exhaust oil mist filter element.

Section 7.0: Recommended Spare Parts Catalog

Keep the following critical replacement items on site to support rapid troubleshooting and prevent prolonged press shutdowns.

Part Number	Description	Primary Section	Factory Stock Status
#TH-2000-K	Type-K Mineral Insulated Thermocouple Sensor Assembly	Section 3.1 (Page 5)	Immediate Shelf Stock
#SSR-75A	75A Solid State Relay with Heatsink Mount	Section 3.1 (Page 5)	Immediate Shelf Stock
#TC-MOD-01	DIN-Rail Multi-Zone Temperature Controller Module	Section 3.1 (Page 5)	Stock / 24 Hours
#HP-2000-P1	Axial Piston Hydraulic Pump Assembly P1	Section 5.3 (Page 8)	3 Business Days
#VALVE-SV2	Proportional Water Solenoid Valve SV-2 (24VDC)	Section 5.3 (Page 8)	Immediate Shelf Stock
#SEAL-KIT-TPP	Main Cylinder Polyurethane High-Temp Seal Kit	Section 4.1 (Page 7)	Stock / 48 Hours

For technical support or emergency hydraulic parts dispatch, contact ThermaForm Industrial Systems Service Division: 1-888-THERMA-PRESS or email support@thermaform-press.com.